



Part-II

Android Application Development CLI Tools

This second article in the series on Android Application Development introduces you to the command line utilities that come packaged with the Android SDK, all of which are extensively used to create an Android Application.

When using an IDE (Integrated Development Environment) like Eclipse, the developer never realises the existence of these utilities, because IDEs provide an abstraction between the developer and the implementation of the process to create an application package. We, being open source enthusiasts and Linux users, should know how to build an application from the command line, and also be aware of exactly what is going on behind the scenes. This gives us a better understanding of the process of creating an Android application and will also help in debugging.

Under *Tools* directory in the folder where the Android SDK is installed there are many command-line utilities. Let's look at those that are used most often in the following sections of this article:

- adb
- android
- ddms
- emulator
- mkshcard
- sqlite3
- traceview



Note: The following invocations/command lines assume that you have added the Android Tools directory to your *SPATH*, and thus do not need to precede each command with an absolute path to its location.

Adb

Adb stands for 'Android Debug Bridge'. *Adb* has a client-server mechanism, where the 'client' is the Android Emulator instance, or the Android device that's connected to the PC, and the 'server' is your development machine (PC). The server communicates with client(s) through a daemon process which runs on the PC. The *adb* server can have multiple clients: for example, two emulator instances and one Android phone connected via USB. To find clients available on the development machine, run the following command:

```
$ adb devices
```

Figure 1 shows sample results of running the command.

Having obtained the list of clients, we can run a command targeting a particular client/instance, by adding the *-s <serial-number>* parameter to the command, where *serial-number* is one of the device names listed by the *adb devices* command. For example:

```
$ adb -s emulator-5554 shell
```

Figure 2 shows the result of running such a command. But to install an Android application into the connected device, the *adb* command given can be one of several, including...

- *install <apk file>*: Installs the given Android